

# Perpendicular lines

The right angle, made and used — and the shortest distance from a point to a line.

LESSON 9–10

2 × 40 MIN

MATEMATIKA 7, §4

S8 5.3

LESSON CLOCK

15'

25'

25'

20'

5'

|                       |        |
|-----------------------|--------|
| Perpendicularity      | 15 min |
| Angles at a crossing  | 25 min |
| The construction      | 25 min |
| The shortest distance | 20 min |
| Homework              | 5 min  |

LEARNING OBJECTIVES

- Define perpendicular lines and use the symbol  $\perp$ .
- Construct a perpendicular through a point on a line and off it.
- Know that the perpendicular is the shortest distance from a point to a line.
- Use vertical and adjacent angles at an intersection.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 26–31          |
| Cambridge | Stage 8, pp. 53–56 |
| Workbook  | Exercise 5.3       |

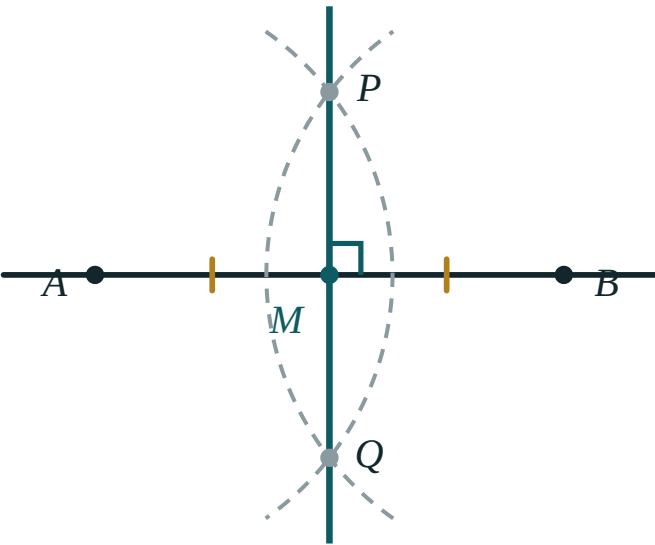
01

## Explanation

### Perpendicularity

Two lines are **perpendicular** if they meet at a right angle. Written  $a \perp b$ .

When two lines cross, four angles are formed. If one of them is  $90^\circ$ , all four are — so one right angle at a crossing makes the lines perpendicular.



Perpendicular lines meet at a right angle; the symbol is a small square.

THROUGH A POINT THERE IS EXACTLY ONE PERPENDICULAR TO A LINE

Whether the point is on the line or off it. This uniqueness is what makes “the distance from a point to a line” a well-defined quantity.

Angles at a crossing

| PAIR     | POSITION                                    | RELATION    |
|----------|---------------------------------------------|-------------|
| adjacent | share an arm, together make a straight line | add to 180° |
| vertical | opposite each other                         | equal       |

**Proof that vertical angles are equal.**  $\angle 1 + \angle 2 = 180^\circ$  and  $\angle 2 + \angle 3 = 180^\circ$ , because each pair is adjacent. Subtracting,  $\angle 1 = \angle 3$ .

| GIVEN                  | THE OTHER THREE                  |
|------------------------|----------------------------------|
| $\angle 1 = 40^\circ$  | $140^\circ, 40^\circ, 140^\circ$ |
| $\angle 1 = 90^\circ$  | $90^\circ, 90^\circ, 90^\circ$   |
| $\angle 1 = 115^\circ$ | $65^\circ, 115^\circ, 65^\circ$  |

ADJACENT MEANS SHARING AN ARM AND MAKING A STRAIGHT LINE

Two angles that merely touch are not adjacent in this sense. The definition matters, because only genuinely adjacent angles are supplementary.

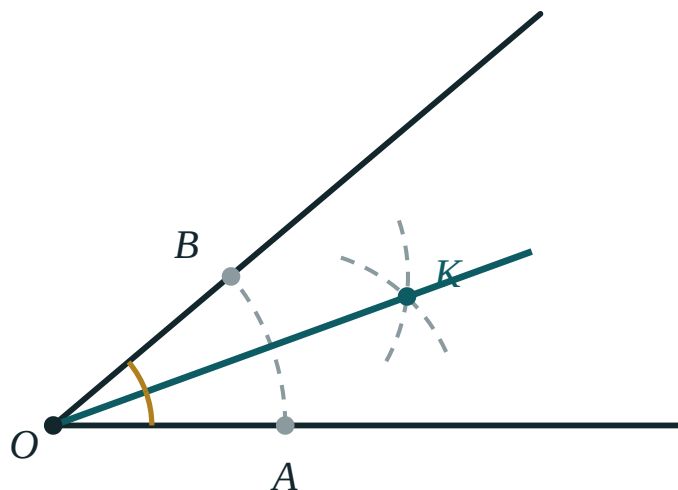
## The construction

### Through a point $A$ on the line $a$ :

1. with centre  $A$ , cut the line at  $P$  and  $Q$  with one arc;
2. with a larger radius, draw arcs from  $P$  and  $Q$  meeting at  $R$ ;
3.  $AR \perp a$ .

### Through a point $B$ not on the line:

1. with centre  $B$ , draw an arc cutting the line at  $P$  and  $Q$ ;
2. from  $P$  and  $Q$ , equal arcs meeting at  $S$  on the far side;
3.  $BS \perp a$ .



*Equal arcs from two points give the perpendicular —  
the same idea in both cases.*

#### BOTH CONSTRUCTIONS ARE THE PERPENDICULAR BISECTOR

In each case  $P$  and  $Q$  are equidistant from the new point, so the line through it is the perpendicular bisector of  $PQ$ . One construction, learnt once.

## The shortest distance

From a point  $B$  off a line, the perpendicular  $BH$  is shorter than every **oblique**  $BC$  to the line.

$BH < BC$  for every  $C \neq H$  on the line

**Why.** In the right triangle  $BHC$ ,  $BC$  is the hypotenuse and  $BH$  a leg — and the hypotenuse is always the longest side.

| TERM                         | MEANING                                |
|------------------------------|----------------------------------------|
| the foot $H$                 | where the perpendicular meets the line |
| the distance from $B$ to $a$ | $BH$                                   |
| the projection of $BC$       | $HC$                                   |

“DISTANCE TO A LINE” ALWAYS MEANS THE PERPENDICULAR DISTANCE

Not the distance to a convenient point on it. This convention is used in every later chapter — in the height of a triangle, in the distance to a plane, in the radius of an inscribed circle.

02 **Worked examples**

EXAMPLE 1 Two lines cross and one angle is  $40^\circ$ . Find the other three.

1

Its adjacent angle:  $180^\circ - 40^\circ = 140^\circ$ .

2

Its vertical angle:  $40^\circ$ .  
Vertical angles are equal.

3

The fourth:  $140^\circ$ .

4

Check:  $40 + 140 + 40 + 140 = 360 \checkmark$

ANSWER

$140^\circ, 40^\circ, 140^\circ$

**EXAMPLE 2** Two adjacent angles are in the ratio 2 : 7. Find both.

① They add to  $180^\circ$ .

② Total parts 9.

③ One part =  $20^\circ$ .

④  $40^\circ$  and  $140^\circ$ .

ANSWER

$40^\circ$  and  $140^\circ$

**EXAMPLE 3** A point  $B$  is 5 cm from a line, and  $C$  on the line is 12 cm from the foot. Find  $BC$ .

①  $\triangle BHC$  is right-angled at  $H$ .

②  $BC^2 = 5^2 + 12^2$   
Pythagoras.

③ = 169

④  $BC = 13$  cm — longer than 5 cm ✓

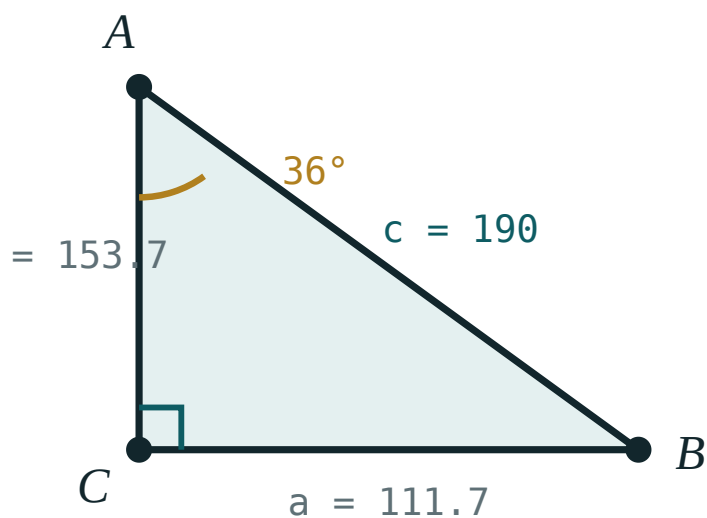
ANSWER

13 cm

### 03 Interactive model

Ask a pupil to walk from a desk to the wall by the shortest route; they walk perpendicular without being told, and the theorem is confirmed before it is proved.

## Perpendicular and oblique

 $\alpha$   $36^\circ$  $\beta$   $54^\circ$ a (opposite) **111.7**b (adjacent) **153.7**c (hyp) **190** $\sin \alpha$  **0.59** $\cos \alpha$  **0.81** $\tan \alpha$  **0.73** $a^2 + b^2$  **36100** $c^2$  **36100**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH                 | O'ZBEKCHA            | РУССКИЙ                  |
|-------------------------|----------------------|--------------------------|
| Perpendicular           | Perpendikulyar       | Перпендикуляр            |
| Right angle             | To'g'ri burchak      | Прямой угол              |
| Foot of a perpendicular | Perpendikulyar asosi | Основание перпендикуляра |
| Distance to a line      | Chiziqqa masofa      | Расстояние до прямой     |
| Adjacent angles         | Qo'shni burchaklar   | Смежные углы             |
| Vertical angles         | Vertikal burchaklar  | Вертикальные углы        |
| Oblique                 | Og'ma                | Наклонная                |
| Set square              | Gonio                | Угольник                 |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$a \perp b$  means they meet at:

45°

90°

180°

any angle

Vertical angles are:

supplementary

equal

complementary

unrelated

Adjacent angles add to:

90°

180°

270°

360°



One angle at a crossing is  $115^\circ$ . The vertical one is:

$65^\circ$

$115^\circ$

$245^\circ$

$90^\circ$

The distance from a point to a line is:

any segment to it

the perpendicular

the longest segment

undefined

Through a point there is:

no perpendicular

exactly one

two

many

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. One angle  $40^\circ$ : its adjacent

ANSWER  $140^\circ$

2. One angle  $40^\circ$ : its vertical

ANSWER  $40^\circ$

3. One angle  $115^\circ$ : its adjacent

ANSWER  $65^\circ$

4. One angle  $90^\circ$ : all four

ANSWER  $90^\circ$

5. Adjacent angles add to

ANSWER  $180^\circ$

6. Vertical angles are

ANSWER Equal

7. Symbol for perpendicular

ANSWER  $\perp$

Medium · the standard question

7 PROBLEMS

1. Adjacent angles in the ratio 2 : 7

ANSWER  $40^\circ$  and  $140^\circ$

2. Adjacent angles in the ratio 1 : 5

ANSWER  $30^\circ$  and  $150^\circ$

3. One adjacent angle exceeds the other by  $40^\circ$

ANSWER  $70^\circ$  and  $110^\circ$

4.  $B$  is 5 from a line,  $HC = 12$ :  $BC$

ANSWER 13

5.  $B$  is 9 from a line,  $HC = 12$ :  $BC$

ANSWER 15

6. Two crossing lines make four angles totalling

ANSWER  $360^\circ$

7. One angle is three times its adjacent

ANSWER  $45^\circ$  and  $135^\circ$

Hard · stretch and reason

7 PROBLEMS

1. The bisectors of two adjacent angles meet at

ANSWER  $90^\circ$

2. Three lines through one point make how many pairs of vertical angles?

ANSWER 3

3.  $BC = 20$  and  $BH = 16$ :  $HC$

ANSWER 12

4. A point 8 from a line: the least possible  $BC$

ANSWER 8

5. Two adjacent angles with bisectors  $90^\circ$  apart: always true?

ANSWER Yes

6. Four angles at a point in the ratio  $1 : 2 : 3 : 4$

ANSWER  $36^\circ, 72^\circ, 108^\circ, 144^\circ$

7. Why is the hypotenuse the longest side?

ANSWER It faces the largest angle

## 07 Homework

### Homework — 5 tasks

Mark every right angle with a small square, and leave construction arcs visible.

- Two lines cross and one angle is  $63^\circ$ . Find the other three.
- Two adjacent angles are in the ratio  $3 : 7$ . Find both.
- Construct the perpendicular to a line through a point not on it.
- A point is  $6\text{ cm}$  from a line; a point on the line is  $8\text{ cm}$  from the foot. Find the distance between them.

5. Explain why the distance from a point to a line is measured along the perpendicular.